PRINCESS NOURAH BINT ABDULRAHMAN UNIVERSITY
College of Computer and Information Sciences Department
of Information Systems



Project name: Hajj and Umrah Analytics with Power BI

Student Name	Student ID
Jood Ahmed Bagazi	*445008562*
Mashael Mohammed Algazla	*445008598*
Reem Alotaibi	*445008577*
Tumadhir Alyahya	*445008599*
Raghad Alqassimi	*445008555*

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Dr. Romana Aziz

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1.0 Introduction

In recent years, the number of pilgrims performing Hajj and Umrah has significantly increased, leading to higher demand on infrastructure, transportation, and crowd management systems. Managing large crowds efficiently is critical to ensure safety, improve pilgrims' experience, and optimize resource allocation.

This project aims to develop an interactive Business Intelligence dashboard using Power BI to analyze Hajj and Umrah data. The dashboard integrates official statistics with crowd behavior data to provide insights into pilgrim distribution, peak periods, and crowd density patterns. Additionally, the project incorporates predictive analytics to forecast future trends and support better decision-making.

2.0 Roles and contribution of team members

Before starting, we all gathered and agreed on the dataset to be used.

Student Name	Work Done
Jood Ahmed Bagazi	Introduction –Problem description – Description of data – Data preparation – Visual model.
Mashael Mohammed Algazla	Dashboard Design and User Interaction
Reem Alotaibi	Dashboard Design
Tumadhir Alyahya	Dashboard Design and User Interaction - Implementation
Raghad Alqassimi	Dashboard Design

3.0 Problem description

The rapid growth in the number of pilgrims creates several challenges related to crowd management, service efficiency, and infrastructure planning. High crowd density can lead to congestion, delays, and potential safety risks, especially during peak seasons such as Hajj and Ramadan.

Currently, data is available in different formats and levels (annual, quarterly, and real-time), making it difficult to analyze and extract meaningful insights. There is a need for a unified system that can combine these datasets and provide a clear understanding of patterns and trends.

The main problem addressed in this project is how to effectively analyze and integrate multiple data sources to identify crowd patterns, predict peak periods, and recommend solutions to improve the overall pilgrim experience.

4.0 Overview of solution

To address the challenges of analyzing Hajj and Umrah data, we utilized Power BI to create intuitive and interactive visualizations. By combining official pilgrim statistics with crowd management datasets, we can identify patterns in crowd density, movement, and peak periods. These insights support better decision-making for resource allocation, crowd control, and service optimization, enabling authorities to improve safety and enhance the overall pilgrim experience.

5.0 Description of data

The data used in this project was collected from two main sources. The first source is the official Hajj and Umrah statistics provided by the General Authority for Statistics in Saudi Arabia. The second source is a crowd management dataset obtained from Kaggle.[1][3]

The official dataset includes information about the number of pilgrims performing Hajj and Umrah. The data is categorized based on origin (Saudi and Non-Saudi), gender (male and

female), and time period (annual and quarterly). This dataset provides an overview of pilgrim distribution and trends over time.

The crowd management dataset contains detailed information about crowd behavior. It includes variables such as crowd density, movement speed, timestamp, location, and environmental conditions. These variables help in understanding how crowds behave in different situations and at different times.

Combining these datasets allows for a comprehensive analysis by linking overall pilgrim numbers with crowd behavior patterns. This supports better understanding of peak periods, congestion levels, and enables predictive insights for improving crowd management and services.

6.0 Data Preparation

Data preparation was performed to clean and structure the datasets before analysis in Power BI. Multiple preprocessing steps were applied to Hajj, Umrah, and crowd management datasets to ensure data quality and consistency.

The following steps were carried out:

- Removed unnecessary rows such as titles and non-data entries
- Renamed columns to meaningful names (e.g., Destination of arrival, Males, Females, Total)
- Converted data types to appropriate formats (e.g., numerical values for counts and ratings, and date/time for timestamps)
- Created new calculated columns:
    - Added a “Year” column to distinguish data across different years
    - Added a “Quarter” column to the Umrah dataset based on the source files (Q1, Q2, Q3, Q4)
    - Extracted the “Quarter” from the timestamp field in the crowd dataset and transformed it into categorical format (Q1, Q2, Q3, Q4) to enable time-based analysis and integration
- Combined multiple datasets:
    - Merged Hajj data from different years into a single table
    - Merged Umrah data from different quarters into a unified table
- Removed irrelevant columns from the crowd dataset to reduce complexity and focus on key analysis variables. The removed columns include:
    - Interaction_Frequency
    - AR_Navigation_Success
    - Nationality
    - Waiting_Time_for_Transport

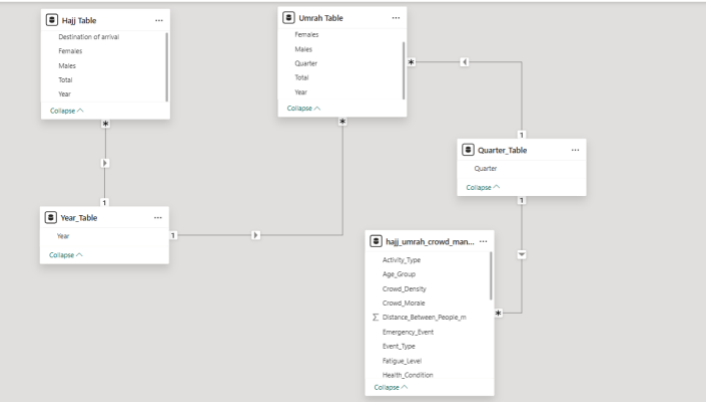
    - AR_System_Interaction
    - Sound_Level_dB
    - Temperature
- Retained important analytical variables such as:
    - Crowd_Density
    - Movement_Speed
    - Fatigue_Level
    - Stress_Level
    - Queue_Time_minutes
    - Distance_Between_People_m
    - Time_Spent_at_Location_minutes
    - Satisfaction_Rating
    - Perceived_Safety_Rating

- Handled missing values (nulls) by filtering out incomplete records from key columns to ensure accurate analysis
- Created supporting dimension tables to improve data modeling and relationships:
    - A Year table was created to manage time-based analysis across Hajj and Umrah datasets
    - A Quarter table was created to link Umrah and crowd datasets efficiently and avoid many-to-many relationships

These steps resulted in a clean, well-structured dataset suitable for visualization, integration, and predictive analysis.

7.0 Visual Model

The model follows a star schema design using dimension tables (Year and Quarter) to avoid many-to-many relationships and ensure acurate filtering across datasets.



[Figure 1:Visual Model]

8.0 Dashboard design overview

8.1 The dashboard is organized into multiple dynamic pages containing key visual elements

The dashboard was designed to present Hajj and Umrah data in a clear and organized way. It was divided into three main pages to make the analysis easier to understand. Each page focuses on a different part of the analysis, including the general overview, crowd analysis, and pilgrim behavior and health analysis.

• Hajj and Umrah Pilgrims Overview: This page provides a general summary of the main Hajj and Umrah statistics. KPI cards were used to show the total number of pilgrims, total males, and total females. A bar chart was used to compare total pilgrims by destination of arrival, while a column chart was used to show yearly pilgrim trends. A donut chart was also added to show the gender distribution of pilgrims.

• Crowd Analysis: This page focuses on crowd-related indicators. It includes visuals that show queue time, activity type, crowd density, stress level, transport mode, incident type, and satisfaction rating. These visuals help users understand how crowd conditions relate to waiting time and stress levels.

• Pilgrim Behavior and Health Analysis: This page focuses on pilgrim behavior and health conditions. It includes visuals related to activity duration, stress level by crowd density, and health condition distribution. This page helps users understand the relationship between pilgrim activities, crowd density, stress level, and health status.

• Slicers:Slicers were added to make the dashboard interactive. Users can filter the data by year, age group, nationality, gender, destination of arrival, or other available fields. This helps users explore specific parts of the data instead of viewing all records at once.

• Navigation Buttons: Navigation buttons were added on the side of the dashboard to make moving between pages easier. This allows users to switch between the overview page, crowd analysis page, and pilgrim behavior and health analysis page smoothly.

8.2. Design and Aesthetic Choices:

• Color Theme: A simple and consistent color theme was used to keep the dashboard clean and professional. The colors were selected to make the visuals easy to read and visually balanced.

• Interactivity: The dashboard includes slicers and filters that allow users to focus on specific data segments, such as year, age group, nationality, gender, and destination of arrival.

• Visual Consistency: Charts and visuals were designed with consistent titles, colors, and formatting. This helps users understand the dashboard without confusion.

• Ease of Use: The dashboard was designed to be simple and easy to navigate. KPI cards were placed at the top to show the most important numbers quickly, while detailed charts were arranged below them in a logical order.

• Clear Layout: The visuals were arranged carefully to avoid crowding. Main indicators were placed at the top, charts were placed in the center, and filters or navigation buttons were placed on the side.

This dashboard helps users understand Hajj and Umrah data through clear visuals, interactive filters, and organized pages. It supports easier analysis by allowing users to compare values, identify trends, and explore crowd and health-related patterns.



Figure 2: Hajj and Umrah Pilgrims Overview Dashboard]



[Figure 3: Crowd Analysis Dashboard]



[Figure 4: Pilgrim Behavior and Health Analysis*]*

9.0 Implementation

9.1 Measures:

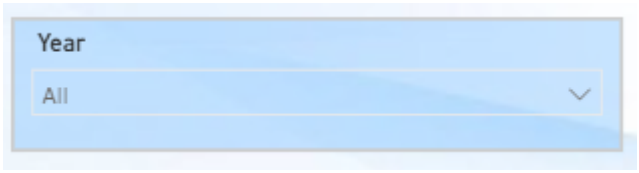
- AVG of QT: Calculates the average queue time (Queue_Time_minutes) for visitors at service points and transportation areas, helping analyze peak hours and crowd congestion
- . Time_Spend_atLocation_inHours: Calculates the total time visitors spend at each location in hours, derived from the Time Spent at Location minutes column, providing insights into activity distribution and crowd density.
- Count: Counts the number of rows in the Crowd Density column from the hajj_umrah_crowd_management_dataset, representing the total number of observations for a specific period or filter.

9.2Dashboard Layout:

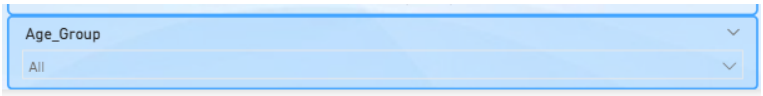
The dashboard title was placed at the top to show the main topic clearly. KPI cards were placed at the top of the page to highlight the most important numbers. Charts were arranged in the center to support comparison and analysis. Slicers and navigation buttons were placed on the side to make the dashboard easier to use.

9.3 Visual

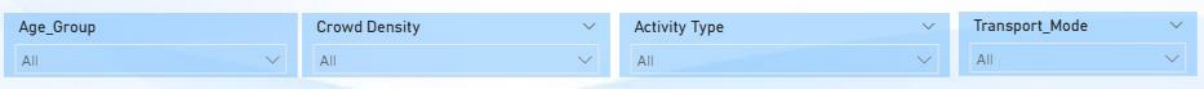
9.3.1 Slicer



*[Figure 5:*Year *slicer in Hajj and Umrah Pilgrims Overview Dashboard]*



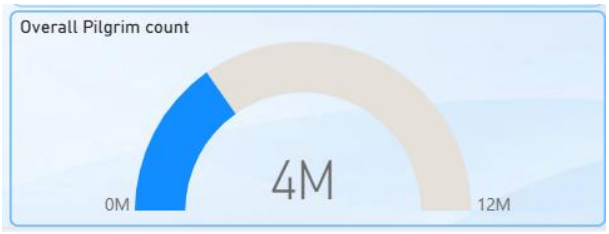
[Figure 6: Age Group slicer *in Pilgrim Behavior and Health Analysis Dashboard]*



[Figure 6: Age Group, Crowd Denesity , Activity, Type, and Transport Mode Slicers *in Crowd AnalysisdDashboard]*

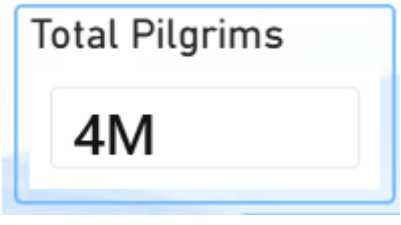
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9.3.2 Gauge Chart

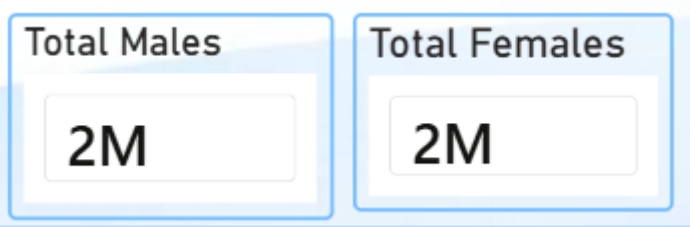


[Figure 7: Overall Pilgrim Count Gauge Chart]

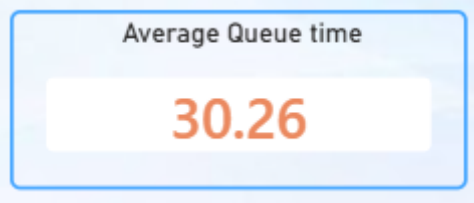
9.3.4 KPI Cards



[Figure 8: Total Pilgrim Cardt]



[Figure 9: Total Males and Females Pilgrim Cards]



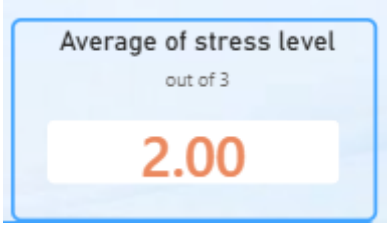
[Figure 10: Average Queue Time Card]



[Figure 11: Average Safety Rating Card]

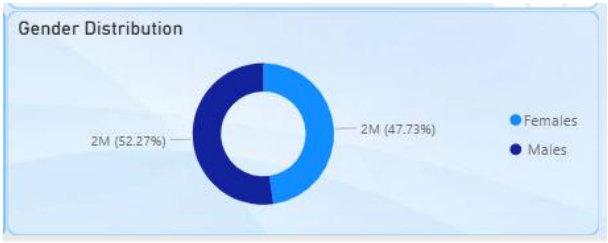


[Figure 12: Average Satisfaction Rating Card]

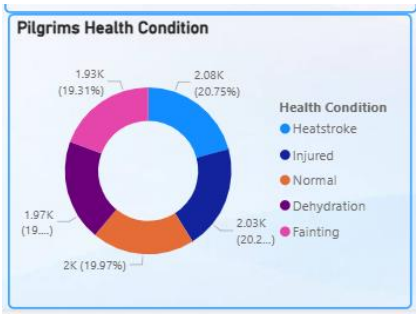


[Figure 13: Average of stress level Card]

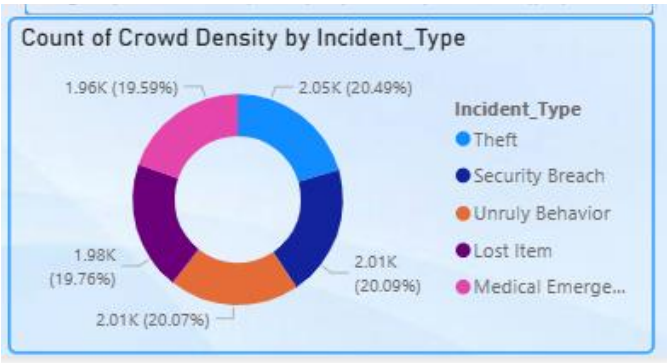
9.3.5 Donut Chart



[Figure 14: Gender Donut Chart]

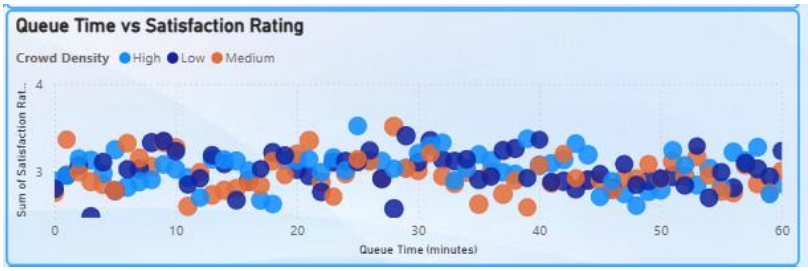


[Figure 15: Pilgrims Health Donut Chart]

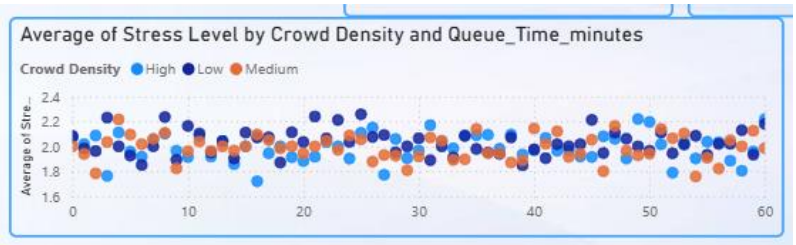


[Figure 16: Crowd Density b7 Incident Type Donut Chart]

9.3.6 Scatter Chart



[Figure 17: Queue Time & Satisfaction Rating Scatter Chart]



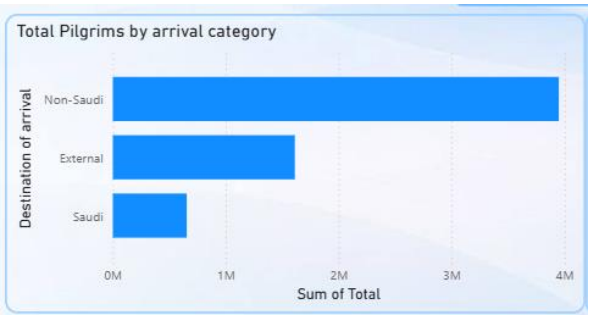
[Figure 18: Average stress level by crowd Density & Queue time in minutes]

9.3.7 Decomposition Tree



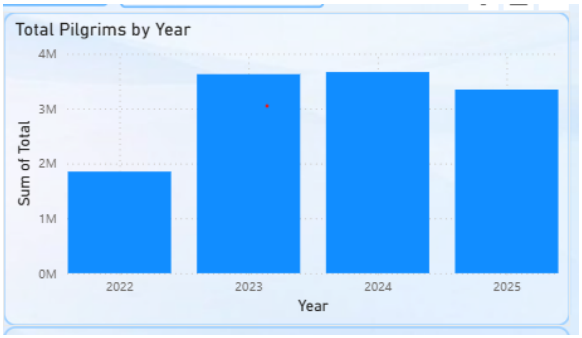
[Figure 19: Stress Level Root Causes Decomposition Tree]

9.3.8 Bar Chart

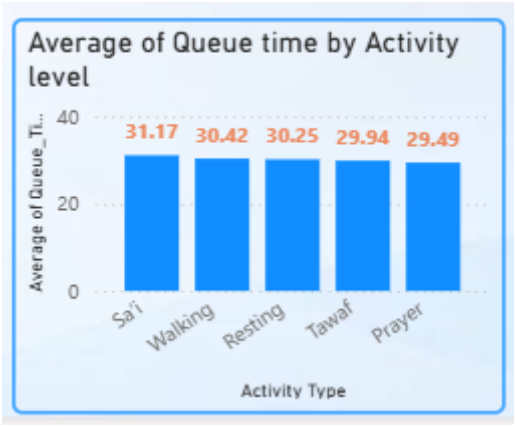


[Figure 20: :Total Pilgrims By Arrival category]

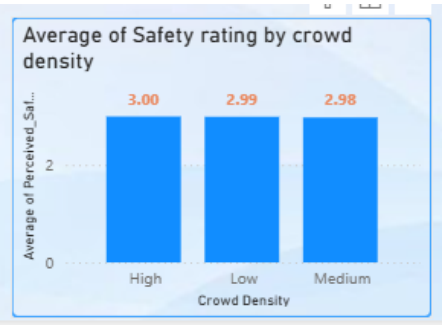
9.3.9 Column Chart



[Figure 21: :Total Pilgrims By year]



[Figure 22: Average of Queue time by Activity level]



[Figure 23: Average of Safety rating by Crow Density]

9.3.10 Button



[Figure 24: Overview, Crowd Analysis, Behavior& Health Analysis Pages Buttons]

10.0 Power BI features used

10.1Visualizations:

Various visual types such as bar charts, column charts, donut charts, and maps were used to represent Hajj and Umrah data, providing meaningful insights into pilgrim distribution, crowd density, and peak periods.

10.2 Interactive Dashboard:

The dashboard is designed to be user-friendly, allowing all users to interact with filters and dropdowns without requiring technical expertise, ensuring smooth exploration of the data.

10.3 Custom Calculations:

Metrics and KPIs were created using DAX to reflect key insights, such as average queue time, time spent at locations, and total pilgrims, reducing dataset complexity and enabling efficient analysis.

10.4 Slicers:

Interactive visual filters allow users to explore data based on year, quarter, gender, nationality, and destination, helping focus on specific conditions and gain targeted insights.

11.0 Problems faced

- Data from different sources had varying formats and levels, making integration challenging.
- .Missing values and inconsistent columns required extensive cleaning
- Designing interactive slicers and navigation for all users required multiple iterations.

- Predictive insights were limited by inconsistent historical data.

12.0 Limitations

Despite the successful development of the Hajj and Umrah Power BI dashboard, the project faces some limitations. Historical data for certain years and locations is limited, which may affect the accuracy of trend analysis and predictive insights. Additionally, some datasets provide only quarterly or annual summaries, restricting the analysis of detailed hourly or daily crowd patterns. Forecasting future crowd behavior is also constrained by the variability of pilgrim numbers and environmental factors not captured in the datasets. Furthermore, the dashboard relies on pre-collected data and does not support real-time updates, and certain metrics such as visitor satisfaction, fatigue, or stress are estimated from available variables, which may not fully represent actual conditions. These limitations should be considered when interpreting the dashboard insights and planning future improvements.

13.0 Future work

enhancements of the Hajj and Umrah Power BI dashboard could focus on several areas. Integrating real-time data feeds from sensors, IoT devices, and crowd monitoring systems would allow live tracking of pilgrim movement and congestion. Advanced forecasting models using machine learning could be implemented to better predict peak periods and optimize resource allocation. Regular updates and improvements based on user feedback will make the dashboard more effective in keeping pilgrims safe, improving services, and enhancing their overall experience.

14.0 Alignment with sustainable development goals

Industry, Innovation, and Infrastructure (SDG 9): The Hajj and Umrah Power BI dashboard contributes to building smarter and more resilient infrastructure at pilgrimage sites. By integrating real-time and historical data, the system enables authorities to monitor crowd movement, congestion, and service utilization efficiently. Predictive analytics and machine learning models help forecast peak periods, optimize resource allocation, and improve traffic and pedestrian flow. IoT devices and sensors provide continuous feedback on crowd density and environmental conditions, supporting rapid decision-making and operational adjustments. These innovations allow for better planning of facilities, transportation networks, and service points, ensuring that infrastructure can handle high volumes of pilgrims safely and sustainably. Overall, the dashboard promotes the development of innovative, efficient, and adaptable infrastructure that enhances both safety and service quality during Hajj and Umrah.[2]

14.0 Conclusions

The Hajj and Umrah Power BI dashboard successfully integrates official statistics and crowd management data to provide a comprehensive, interactive platform for analyzing pilgrim patterns. The dashboard enables authorities to monitor crowd density, movement, queue times, and safety indicators, supporting informed decision-making for resource allocation, infrastructure planning, and service optimization. Predictive analytics and interactive visualizations allow stakeholders to identify peak periods, anticipate congestion, and implement preventive measures, enhancing overall pilgrim experience and safety. By aligning the dashboard with Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure), the project contributes to developing smart, resilient, and efficient infrastructure at pilgrimage sites. Overall, this solution demonstrates the power of business intelligence in improving operational efficiency, safety, and service quality for large-scale religious events

15.0 References

[1] General Authority for Statistics, Saudi Arabia, "Hajj and Umrah Statistics," 2026. [Online]. Available: https://www.stats.gov.sa

[2] United Nations, "Sustainable Development Goal 9: Industry, Innovation and Infrastructure," 2023. [Online]. Available: https://sdgs.un.org/goals/goal9

[3] Z. Ziya, “Hajj and Umrah Crowd Management Dataset,” Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/ziya07/hajj-and-umrah-crowd-management-dataset

Appendices:

A. Readme File – Report
B. Readme File - Dashboard